Abu Dhabi City Campus Daily/Weekly Planner

Week 11: 9th – 13th November 2026

Subject: Mathematics **Unit Title: Differentiation** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: **Basic Derivative Rules**	I can differentiate polynomial functions using the constant, sum, difference and constant multiple rules.	2.1 Clarify derivative rules, symbols and algebraic vocabulary using annotated examples and colour-coded steps. 2.3 Support decoding of polynomial notation through expanded and factored representations.	Related Videos / Resources Basic Derivatives Examples | Calculus Basic Derivatives Examples | Calculus Using Derivatives to Find Slope & Tangent Lines | Calculus Lesson 10
Lesson 2: **Applications of Derivative Rules**	I can combine basic differentiation rules to find the derivative of polynomial expressions and interpret the result as a rate of change.	2.3 Connect algebraic differentiation with graphical and verbal interpretations. 5.3 Build fluency through guided examples followed by independent multi-step practice.	
Lesson 3: **Equation of a Tangent Line**	I can determine the gradient of a tangent to a curve at a given point and use it to construct the equation of the tangent line.	5.3 Build fluency by progressing from finding f'(a) to constructing a tangent-line equation. 8.2 Vary challenge through curves with positive, negative and zero tangent gradients.	
Lesson 4: **Unit 2 - Formative 1**	I can demonstrate my understanding of the definition of a derivative, the power rule and basic differentiation rules through mathematical problems.	8.2 Vary challenge level through a range of foundational, standard and higher-order assessment questions.	
Homework	Complete assigned practice on constant, sum, difference and constant multiple rules; applications of differentiation; and tangent-line equations. Worksheets are available on Teams.		
Teacher's notes/ Reminders	Students should bring a GDC/graphing calculator, notebook and device. Ensure students can access lesson materials on Teams/OneNote.		

Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator; Desmos; lesson PowerPoints; digital practice and assessment materials.
Cross-curricular links	**Physics/Science:** Interpret derivatives as instantaneous rates of change, including velocity from displacement and rates of changing physical quantities. **Engineering/Design:** Use tangent gradients to estimate local behaviour of curves and construct linear approximations.
External Assessment Integration	Direct alignment with AP Calculus AB Unit 2: differentiation skills, derivative notation, polynomial differentiation and tangent-line applications. Include AP-style multiple-choice and free-response practice.
Wellbeing/ ATL Skills	**Thinking Skills:** Select and apply appropriate differentiation rules; connect algebraic, graphical and contextual representations; analyse errors and justify results. **Communication Skills:** Use accurate derivative notation, mathematical vocabulary and clear working.

	Self-Management Skills: Organise multi-step solutions, practise independently, manage assessment time and reflect on feedback.
Standards	AP Calculus AB Unit 2: Differentiation Defining the Derivative Basic Differentiation Rules (constant, sum, difference and constant multiple rules) Applications of derivative rules Tangent lines Rates of change
Inquiry/ TOK Question	How does the derivative help us describe how a quantity changes at a particular instant, and how can that information be used to construct a tangent line?
Success Criteria (Measurable)	**Lesson 1: Basic Derivative Rules** **All students must** differentiate constant and polynomial terms using the constant and power rules, and apply sum, difference and constant multiple rules in straightforward examples. **Most students should** differentiate multi-term polynomial functions accurately and show clear algebraic working. **Some students could** select and combine the appropriate rules in unfamiliar polynomial expressions and explain why each rule applies. **Lesson 2: Applications of Derivative Rules** **All students must** apply basic differentiation rules to find the derivative of a polynomial expression. **Most students should** combine two or more rules accurately and interpret the derivative value as an instantaneous rate of change in a given context. **Some students could** analyse an unfamiliar contextual model, calculate its derivative and explain the meaning and units of the result. **Lesson 3: Equation of a Tangent Line** **All students must** find the derivative and evaluate it at a specified x-value to determine the tangent gradient. **Most students should** use the point-slope form and the given point on the curve to construct the tangent-line equation. **Some students could** solve multi-step tangent-line problems, verify that the line passes through the point of tangency and interpret the gradient in context. **Lesson 4: Formative Assessment 1 – Unit 2** **All students must** demonstrate understanding of the derivative definition, power rule and basic differentiation rules in routine questions. **Most students should** solve multi-step differentiation problems accurately and communicate mathematical reasoning using correct notation. **Some students could** solve unfamiliar or higher-order problems that require selecting and combining differentiation methods and justifying their approach.
National Identity Domain	**Respect** الاحترام
	Area: Citizenship
	Implementation: Promote mathematical responsibility by showing complete working, using precise derivative notation and checking the reasonableness of calculated gradients and tangent-line equations.